

The Golden Threshold for Greedy Packing of Nested Rings

Javier Aguilar Martín
AGILabs

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Abstract

We prove the golden threshold for descending greedy packing of nested rings in a disk. The geometric ingredient is a three-largest-disks theorem: under golden tail bounds, a finite list of disks fits a circular container if and only if its three largest members fit. This supplies two auxiliary slots at every branching container and closes a parent-preserving exchange for an arbitrary finite inventory. The argument permits independent hole radii. Combined with the four-ring counterexamples approaching the golden ratio from above, the lower bound gives an unattained threshold $\tau = \varphi$. Seven new Lean certificates verify algebraic steps; the Euclidean geometry and forest assembly remain written proofs.

1 Model and statement

Let $r_1 > \dots > r_n > 0$ be outer radii. Ring i has a concentric hole of radius $h_i \in [0, r_i]$; its hole may be chosen independently of the others. Common width $w > 0$ is the special case $h_i = \max(r_i - w, 0)$. The pan is a disk of radius R . A placement is specified by a rooted forest, with the pan as a virtual root: the parent of each ring is the pan or a larger ring, and the outer disks of the immediate children of a container must pack inside it with disjoint interiors. Tangencies are allowed. Local packings assemble by translation along the forest.

A descending greedy execution examines the radii in order. It accepts a ring whenever at least one current container can accommodate it with its existing children, and chooses any such container. Siblings may be repacked; the parents of previously accepted rings remain fixed. Let

$$\varphi = \frac{1 + \sqrt{5}}{2}, \quad \rho = \max_{1 \leq i \leq n} \frac{\sum_{j>i} r_j}{r_i}, \quad \beta(a, b) = \frac{ab(a + b)}{a^2 + ab + b^2}.$$

Lexicographic order on feasible subinventories gives priority to including the largest radius at which they differ.

Theorem 1 (Uniform golden guarantee). *If $\rho \leq \varphi$, every descending greedy execution returns the lexicographically maximum feasible subinventory. This holds for every finite inventory in the plane, including independent hole radii.*

The proof below does not bound n , nor enumerate inventories of successive sizes. The induction parameter is the number of already aligned parents in an arbitrary fixed finite inventory.

2 Tail bounds and a geometric pocket

A list of disks whose radii sum to s fits in a disk of radius s : place them consecutively along a diameter. This row construction may replace a disk envelope without changing its surrounding packing.

Lemma 2 (Two tail bounds). *If $p + U \leq \varphi m$ and $U \leq \varphi p$, then $U \leq m$. If $a > b > 0$, $T \leq \varphi b$, and $b + T \leq \varphi a$, then $T \leq 2\beta(a, b)$.*

Proof. For the first assertion, $(1 + \varphi)U \leq \varphi p + \varphi U \leq \varphi^2 m = (1 + \varphi)m$. For the second, normalize $b = 1$, put $x = a/b > 1$, $g = x(x+1)/(x^2+x+1)$, and write $t = T/b$. We have $t \leq \min(\varphi, \varphi x - 1)$. The identities

$$\begin{aligned} 2x(x+1) - \varphi(x^2 + x + 1) &= (x - \varphi)((2 - \varphi)x + 1), \\ (x^2 + x + 1) + 2x(x+1) - \varphi x(x^2 + x + 1) &= (\varphi - x)(\varphi x^2 + (2\varphi - 2)x + \varphi - 1) \end{aligned}$$

give $2g \geq \varphi$ when $x \geq \varphi$, and $1 + 2g \geq \varphi x$ when $1 < x \leq \varphi$. Thus $t \leq 2g$ in both cases. Rescale by b . $\square$

Lemma 3 (Balanced rows). *Let a nonempty finite list of positive radii have total T and largest member p . After reserving p , its remaining members can be divided into two groups, each of total at most $T/2$.*

Proof. Assign each remaining member to the currently lighter group. Both load differences remain at most p , since the added member is at most p . At the end the loads u, v satisfy $u + v = T - p$ and $|u - v| \leq p$; hence $2u, 2v \leq T$. Each group fits its row envelope. $\square$

Place disks $a \geq b > 0$ in a disk of radius $a + b$, with centers $(-b, 0)$ and $(a, 0)$. The upper and lower pockets have inscribed disks of radius $\beta(a, b)$, each tangent to the two disks and the wall. Their centers are

$$\left(\frac{(a-b)(a+b)^2}{a^2 + ab + b^2}, \pm 2\beta(a, b) \right).$$

These coordinates verify all three tangencies and separation of the two pockets directly. We need a stronger fact about a single pocket.

Lemma 4 (Two unequal disks in one pocket). *If $0 < r, s \leq \beta(a, b)$ and $r + s \leq b$, the two disks of radii r, s fit simultaneously in the upper pocket. The lower pocket remains available. A zero radius may be omitted.*

Proof. Put $L = a + b$. A disk of radius $x \leq \beta(a, b)$ tangent to the wall has its center in the upper half-plane at an admissible angle exactly in

$$\left[2 \arcsin \sqrt{\frac{bx}{a(L-x)}}, \pi - 2 \arcsin \sqrt{\frac{ax}{b(L-x)}} \right].$$

The law of cosines gives the two endpoints. The interval is nonempty exactly when $x(a^2 + b^2) \leq ab(L - x)$, equivalent to $x \leq \beta(a, b)$. Place r at its lower endpoint and s at its upper endpoint, and set

$$A^2 = \frac{as}{b(L-s)}, \quad B^2 = \frac{br}{a(L-r)}, \quad A, B \geq 0.$$

Their angular separation is $\pi - 2 \arcsin A - 2 \arcsin B$; the required separation is $2 \arcsin(AB)$. It suffices that $A^2 + B^2 + 3A^2B^2 \leq 1$: this implies $\sqrt{1-A^2}\sqrt{1-B^2} \geq 2AB$ and therefore $\arcsin A + \arcsin B + \arcsin(AB) \leq \pi/2$. After multiplication by the positive denominator $ab(L-r)(L-s)$, the margin is

$$L^2[a(b-r-s) + (a-b)r] + (a-b)^2rs \geq 0.$$

The disks lie wholly above the diameter: its intersection with the container is covered by a, b , so a disk disjoint from them cannot cross it. The lower pocket is consequently free. $\square$

3 The minimal triple and its opposite gap

Lemma 5 (A fully tangent minimal triple). *For $a \geq b \geq c > \beta(a, b)$, the three disks admit a smallest containing disk of radius $R_0 > a + b$, in which they are pairwise tangent and all tangent to the wall. With*

$$A = 1/a, \quad B = 1/b, \quad C = 1/c, \quad t = \sqrt{AB + AC + BC},$$

its radius satisfies $1/R_0 = 2t - A - B - C$.

Proof. Existence of a minimum follows by compactness, using a row as an upper bound. In any feasible placement of three disks, each disk can be moved to the wall while the other two are fixed. For its center x , choose a nonzero direction v such that $(x - x_j) \cdot v \geq 0$ for the other two centers. Two closed half-planes through the origin have such a common direction in the plane. Distances to both other centers are nondecreasing along the ray. Move to the wall and repeat for the others.

For wall-tangent disks with pair sums at most R , let $\theta_{ij} \in [0, \pi]$ denote their minimum angular separation. Three such positions exist exactly when

$$\theta_{ab} + \theta_{ac} + \theta_{bc} \leq 2\pi.$$

Necessity follows by summing the three directed cyclic gaps, each at least its corresponding minimum. Conversely choose gaps in $[\theta_{ij}, \pi]$ summing to 2π ; their endpoints have sums at most 2π and 3π , so this is possible. Each gap is then the smaller angle for its pair.

Two-disk necessity gives $R_0 \geq a + b$. At equality the two largest are forced into the diametral arrangement. Move c to the wall while fixing them and use the interval in Lemma 4; it would give $c \leq \beta(a, b)$, a contradiction. Thus $R_0 > a + b$. All three minimum angles are continuous and strictly between 0 and π near R_0 . Their sum equals 2π , since a strict inequality would permit decreasing R_0 . Taking these gaps realizes every tangency. Descartes' equation for the oriented curvatures $A, B, C, -1/R_0$ gives the stated formula; its other root has $1/R_0 < 0$. $\square$

Lemma 6 (Compensation by the other gap). *In the placement of Lemma 5, there is a further disk of radius $d > 0$, disjoint from all three disks and contained in the same container, with $c + d \geq 2\beta(a, b)$.*

Proof. First justify the geometric existence. Invert about the tangency point of a, b . Their boundaries become parallel lines and their interiors become the exterior half-planes of a strip. The container interior becomes the exterior of a circle inscribed in the strip. A disk inside the strip tangent to both lines has radius equal to half its width; its center lies on the midline.

Tangency to the image of the container leaves exactly two such disks, on opposite sides of that circle. They have disjoint interiors. The inversion center lies inside the image of the container circle, hence outside both solution disks; their inverse images are bounded disks. Inverting back gives two admissible disks in the original container, one being c and the other denoted d . This argument identifies the two components of the free region; a height inequality alone would not establish their disjointness.

Set $S = A + B$, $P = AB$, and $Q = S^2 - P$. Then $t^2 = P + SC$ and $c > \beta(a, b) = S/Q$ imply $SC < Q$, hence $t < S$ and $C < S$. The classical Descartes reflection formula for the two solutions tangent to a, b and the wall [1, Section 3.3] gives

$$D = \frac{1}{d} = 2(A + B - 1/R_0) - C = 4S + C - 4t > C > 0.$$

The roots are distinct because $t < S$, and inversion has exhibited exactly two admissible disks, so the reflection identifies the other one. Using $t^2 = P + SC$ and the displayed expression for D yields

$$Q(C + D) - 2SCD = 2(S - t)^2(2S + 2t - C) \geq 0.$$

Divide by $QCD > 0$ to obtain $c + d \geq 2S/Q = 2\beta(a, b)$. $\square$

4 Three largest disks determine local feasibility

Theorem 7 (Three largest disks). *Let $a > b \geq c \geq r_4 \geq \dots \geq r_n > 0$, $n \geq 3$, and suppose every tail is at most φ times its corresponding radius. The entire list fits a disk of radius R if and only if a, b, c fit. Equality between the smaller radii is permitted.*

Proof. Only sufficiency needs proof. Let $U = \sum_{j \geq 4} r_j$. Lemma 2, applied to a, b , gives $c + U \leq 2\beta(a, b)$.

If $c > \beta(a, b)$, place the three largest disks in their minimum container $R_0 \leq R$. Lemma 6 provides a disk of radius $d \geq 2\beta(a, b) - c \geq U$ in the remaining gap. Replace it by a row containing all the remaining disks.

If $c \leq \beta(a, b)$, place a, b diametrically in radius $a + b \leq R$ and place c in the lower pocket. For a nonempty remaining list, write $e = r_4$ and $V = \sum_{j \geq 5} r_j$. The two-tail assertion of Lemma 2, first at b, c and then at c, e , gives

$$e + V \leq b, \quad V \leq c, \quad e \leq c \leq \beta(a, b).$$

Thus Lemma 4 places the two envelope disks e, V in the upper pocket. Replace V by a row of its actual members; omit it when $V = 0$. The empty remaining list requires no extra step. These envelopes construct positions; they do not replace the inventory in the definition of ρ or in a greedy execution. $\square$

5 A uniform exchange of parents

Suppose a descending execution differs from the lexicographic maximum. At the first differing radius it must omit a ring of that maximum: the opposite difference would contradict lexicographic maximality of the latter. Restrict to the already admitted set G and this omitted ring. This restricted inventory is feasible and still has $\rho \leq \varphi$. Let F be its greedy forest on G . Among

all complete witness forests P , choose one having the longest prefix of parents in agreement with F ; this maximum exists because there are finitely many parent assignments. If all parents agree, the omitted ring can be accepted in F , contradicting rejection.

Otherwise let m be the largest disagreement, with destination u in F and origin $v \neq u$ in P . Both containers are the root or holes of rings larger than m . All larger parents agree. All smaller rings may be reconstructed as leaves, since we only need to align the prefix through m .

Write T for the total radius below m . If $T \leq m$, remove the smaller rings, put m in u using the prefix certificate from F , and place all smaller rings in a row inside the outer disk vacated by m in v , preserving the larger siblings there. This aligns m . Hence assume $T > m$. Let $p < m$ be the next radius and $U = T - p$. The first assertion of Lemma 2 gives $U \leq m$.

Consider the rooted tree formed by the larger rings and the virtual root. The degree of a container is its number of immediate larger children. The following three cases cover every such tree.

A branching container with at least three larger children. Let w have children $a_1 > \dots > a_k > m$, $k \geq 3$. The auxiliary disk list

$$a_1, \dots, a_k, m, m$$

satisfies all golden tail bounds: for each a_i its auxiliary tail is at most its original tail, because $2m < m + T$; the last two tail ratios are 1, 0. Its three largest disks already coexist in w . Theorem 7 therefore gives two radius- m slots alongside all the larger children.

If $w \neq u$, put p in one slot, a row of total U in the other, and m in u using F . This includes $w = v$; no third slot in w is needed. If $w = u$, put m, p in its slots and the row of total U in the outer disk vacated by m in v , using P there.

A container with exactly two larger children. Let its children be $a > b > m$. Their original tails imply $m + T \leq \varphi b$ and $b + m + T \leq \varphi a$; thus $m + T \leq 2\beta(a, b)$ by Lemma 2. Since $T > m$, $\beta(a, b) > m$. The container has radius at least $a + b$, so a diametral placement of a, b provides two radius- m slots in its opposite pockets. Apply the preceding slot assignment. This also proves the degree-two criterion without a separate case enumeration.

No branching container. The larger tree is a chain with exactly one leaf container. Since $u \neq v$, at least one of u, v has exactly one larger child a . If it is u , F certifies the pair a, m there; if it is v , P does so. In either case that container has radius at least $a + m$. The original tails give $T \leq \varphi m$ and $m + T \leq \varphi a$, hence $T \leq 2\beta(a, m)$. Reserve p ; Lemma 3 divides the others into two rows of total at most $T/2 \leq \beta(a, m)$.

When using u , put a, m there diametrically, the two rows in its pockets, and p in the outer disk vacated by m in v . When using v , keep a there, put p inside the envelope reserved for a phantom disk m , and put the two rows in the pockets; the real m goes to u using F .

In every case the parents of all larger rings are unchanged. Local interior packings move with their owners. Those owners have radius larger than m , so the assembly creates no cycle even when one chosen container is an ancestor of another. The complete new witness agrees with F through m , contrary to the maximal choice of P . This proves Theorem 1.

6 The exact threshold and verification scope

Let τ be the infimum of ρ among common-positive-width disk instances for which some descending greedy execution misses the lexicographic maximum. The golden four-ring family already established in the companion manuscript [2, the theorem labelled `thm:golden` in the source] has

$$R = \varphi + 1, \quad w = 0.3, \quad r = \left(\varphi, 1, \frac{\varphi}{2} + 2\varepsilon, \frac{\varphi}{2} + \varepsilon \right)$$

for sufficiently small $\varepsilon > 0$, with $\rho = \varphi + 3\varepsilon$. All four rings fit, but placing the first two in the pan forces a rejection later. Consequently $\tau \leq \varphi$.

Corollary 8. *Together with that previously proved upper bound, Theorem 1 gives $\tau = \varphi$, and the infimum is not attained. The same threshold holds for the class allowing independent hole radii, since that class contains the common-width family.*

Formal and independent checks. The module `Calamares/ThreeCore.lean` adds seven algebraic certificates: the angular margin identity and its sign, the gap-sum identity and its sign, an auxiliary height sign, the two envelope bounds, and the passage from curvature to radius inequalities. Together with the existing tail and partition certificates, they compile in Lean 4.32.2 core. The repository has 122 theorem declarations, without admissions or new axioms. These certificates do not formalize inversion, angular placement, the minimum-container argument, or forest assembly.

The deterministic script `code/tres_mayores.py` constructs centers and separately checks all walls and disk pairs on 10 000 random lists and six boundary examples. It uses no optimizer; these floating-point checks are not a universal proof. Claude Code/Fable performed two static reviews of the Spanish source proof and its dependencies. The final review accepted the complete universal guarantee, with clarifications incorporated here; it did not execute Lean or Python. The English exposition in this supplement was prepared locally from that reviewed proof, and was not separately submitted to Fable. The source review packets, verdicts and validation records are preserved in `docs/reviews/`. This supplement does not replace the frozen v2 manuscript or its reference bundle.

References

- [1] R. L. Graham, J. C. Lagarias, C. L. Mallows, A. R. Wilks, and C. H. Yan, *Apollonian Circle Packings: Geometry and Group Theory I. The Apollonian Group*, Discrete Comput. Geom. 34 (2005), 547–585. <https://arxiv.org/abs/math/0010298>.
- [2] J. Aguilar Martín, *Greedy Packing of Nested Rings: Placement Rules, a Golden Counterexample, and a Tribonacci Floor*, v2 working manuscript, 14 September 2026. Source: `paper/main.tex`, accompanying repository <https://github.com/JaviMaligno/calamares>.